



Nalinaksh VYAS, 52

INNOVATION

**Nano satellite
called JUGNU**

COST OF PRODUCTION

Rs 3 crore

WHAT IT WILL DO

**Monitoring droughts,
urban planning and
wasteland management**

big satellites do, also pointing out that the Anna University, Tamil Nadu, had earlier developed a 40-kg micro-satellite. Satellites weighing up to 10 kg are called nano satellites while those in the 10-100 kg range are micro-satellites. Agrawal says that the idea to develop a nano satellite was initiated during the 2007 convocation by the then ISRO chief Madhavan Nair. Finally an MOU was signed between IIT-K and ISRO in January 2009. Developing Jugnu has cost IIT-K approximately Rs 3 crore.

The team worked 16 hours a day, even skipping Sundays and holidays. "We are really excited about the project as it would reinforce IIT-K's image

A TEAM OF 45 IIT-K STUDENTS ASSISTED THEIR PROFESSORS IN MAKING INDIA'S FIRST NANO SATELLITE FOR ISRO.

**VYAS AND HIS YOUNG
RESEARCHERS IN
THE IIT-K LABORATORY**

in the field of space research", says Saket Saurav, 21, who worked on the integration part of the project. Many students remained holed up in their hostel rooms for months on end. "When we pulled them out, they were in a shabby shape with long beards and unkempt hair," laughs Vyas.

Jugnu weighs just 3.5 kg, is 10 cm x 10 cm x 32 cm and has most functionalities of a normal satellite. The payload of the satellite will include an indigenously designed camera for near-infra red remote sensing, a GPS receiver and a micro-electro-mechanical system-based inertial measurement unit. Jugnu would also transmit a blinking signal at all times. Amateur frequency bands would be used for communication so that Jugnu can be tracked by the IIRAM radio community anywhere in the world.

Vyas says though the life of a nano satellite is six months, "we are optimistic that Jugnu can remain in the polar orbit for up to a year." After its launch, Jugnu would be controlled by the ground station at the IIT-K campus. Made mostly of aluminum alloy and electronics, it would be in a polar orbit at an altitude of about 700 km above the earth's surface with a 100-minute orbit-time and would take 14-15 rounds of the earth daily. Scientists said that though it has numerous applications, its high resolution pictures can be used for drought monitoring, wasteland management, urban planning and flood risk management.

Why Jugnu assumes significance compared to other satellites is not only because of its small size but also because it will become a platform for testing the latest technologies in space research. Jugnu comprises 11 subsystems, six housekeeping subsystems and three payloads. Additionally another subsystem, the ejection system, has been indigenously designed for the separation of the satellite from the launch vehicle. The IIT-K team has been assisted by a team of ISRO scientists led by the director, Small Satellite.

Once launched, it would push the limits in space research. Jugnu the midget showcases the giant leap IIT-K has taken in space research. ■